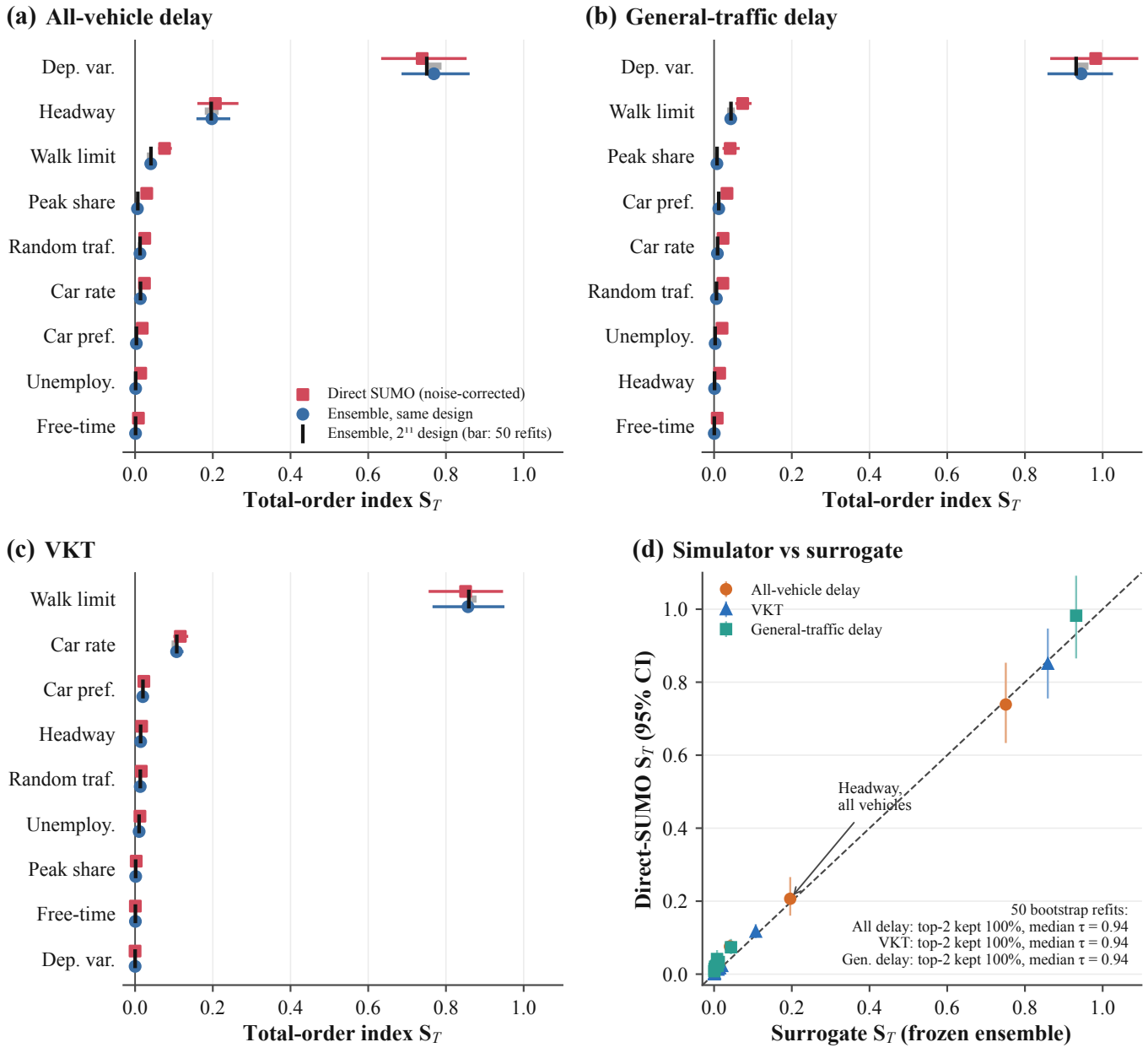

(a) All-vehicle delay
Dep. var.
Headway
Walk limit
Peak share
Random traf.
Car rate
Car pref.
Unemploy.
Free-time
Direct SUMO (noise-corrected)
Ensemble, same design
Ensemble, 2^11 design (bar: 50 refits)
0.0
0.2
0.4
0.6
0.8
1.0
Total-order index S_T
(b) General-traffic delay
Dep. var.
Walk limit
Peak share
Car pref.
Car rate
Random traf.
Unemploy.
Headway
Free-time
0.0
0.2
0.4
0.6
0.8
1.0
Total-order index S_T
(c) VKT
Walk limit
Car rate
Car pref.
Headway
Random traf.
Unemploy.
Peak share
Free-time
Dep. var.
0.0
0.2
0.4
0.6
0.8
1.0
Total-order index S_T
(d) Simulator vs surrogate
All-vehicle delay
VKT
General-traffic delay
Headway, all vehicles
50 bootstrap refits:
All delay: top-2 kept 100%, median τ = 0.94
VKT: top-2 kept 100%, median τ = 0.94
Gen. delay: top-2 kept 100%, median τ = 0.94
Direct-SUMO S_T (95% CI)
0.0
0.2
0.4
0.6
0.8
1.0
Surrogate S_T (frozen ensemble)